# AUtoMLy | Medicaid Policy Copilot

## INAI 6986 — Milestone 2: Model Testing, Validation and Refinement

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**Result: the fine-tuned model does not meet the thresholds**

The one-time final test ran on October 19 on 150 fresh questions. With the source passage supplied, the untouched base model was as accurate as my fine-tuned adapter on answerable questions, and the adapter almost never declined questions the source cannot answer. None of the four measured thresholds is met. I do not recommend deployment.

## 1. Summary of Results

The baseline is Nemotron 3.5 Lightning's untouched base model; the model is the same base with my fine-tuned adapter (1,600 training records, checkpoint 400). Both received the same official source passage and question. On the 120 answerable questions the base gave 114 fully correct, supported answers (95.0%) and the adapter 111 (92.5%): a difference of -2.5 points against my threshold of at least +10, and not statistically reliable ($p = 0.508$). The sharper failure is on the 30 questions whose answer is not in the passage: the base correctly declined 24 (80%), the adapter only 2 (6.7%), against a threshold of 90%. The adapter put an unsupported claim in 25 of 150 answers (16.7%; base 4.0%), against a threshold of under 2%, and named the correct page in only 11 of 120 answers. In business terms, per 100 questions whose answer is in the source, either model gives about 93 to 95 usable drafts; per 100 questions whose answer is not in the source, the adapter would still produce a confident answer about 93 times, the base about 20. Verdict: the model does not meet the threshold and would not be deployed; the untouched base model with the passage supplied is the better drafting aid.

## 2. Experimental Setup

**Frozen configuration.** NVIDIA Nemotron 3.5 Lightning 30B-A3B, BF16, revision a9904d2. Adapter: LoRA rank 16 trained on 1,600 records, checkpoint 400 (adapter SHA-256 4aaca5c5…). Decoding: greedy (no sampling), at most 512 new tokens, reasoning mode off, seed 20261006, one question at a time on one RTX PRO 6000 Blackwell GPU. The base condition is the same loaded weights with the adapter switched off. Development runs gave the model only the question. For the final test, both conditions receive the same official source passage with the question, so the test measures whether fine-tuning improves reading and citing the right rule rather than recalling a manual from memory. No retrieval system is involved.

**Split and test protection.** Records are split by document (1,684 train / 124 validation / 191 test). All tuning used validation records only. The 191 historical test records were used for diagnostics in September, so I no longer treat them as untouched. The final test used 150 fresh questions (120 answerable, 30 unanswerable) from ten official documents outside all three splits that an October 14 audit cleared; one flagged pharmacy chapter was excluded. OpenAI GPT-6.1 Sol wrote the questions and reference answers from the source pages, and code checked every answer quote and page locator against the page text. I did not write the questions; the benchmark is AI-written and source-checked. Membership, prompts, settings and scoring rules were sealed and committed on October 19 before the first scored output, and nothing was tuned afterward. A single-item smoke test confirmed that the container loads and does not count.

**Metrics.** Primary: strict accuracy on answerable questions. An answer counts only if it is correct, supported by the source passage and free of unsupported claims. Secondary: unsupported-claim rate, correct abstention on unanswerable questions (a pass means declining without inventing the missing fact) and citation correctness, checked in code (the answer must name the expected document and page). These reflect the business asymmetry: a confident wrong policy statement can lead to a wrong billing decision, while a refusal only costs analyst time.

**Changes from the PM Plan, declared before the test.** (1) The PM Plan committed to human-judged accuracy. The final test keeps a human check but changes how it is done: two LLM judges, OpenAI GPT-6 Luna and Anthropic Claude Sonnet 5.5, first scored every output separately without seeing each other's ratings or which model wrote the answer. I then reviewed their ratings: for the main run (300 outputs) every rating, and for the 60%-cut run only the outputs where the judges disagreed. Where the judges disagreed I chose the rating I judged correct; where they agreed, their rating stood unless I flagged a problem. On October 19, before I opened the review sheet, I simplified it so that I recorded approval once and marked only disagreements and problems. I flagged none, so every final rating is one of the two judges' ratings. Both judges passed a synthetic check before touching final data. Neither access route allows setting a temperature, so I do not claim the scores are deterministic. (2) The October development judge matched my review on correctness, support and unsupported claims but on citation validity in only 3 of 12 shared cases, so code checks citation identifiers.

**Final-test protocol, sealed before the single run**

| Element | What was fixed in advance |
|---|---|
| Benchmark | 150 fresh questions (120 answerable, 30 unanswerable) from ten official documents, written by GPT-6.1 Sol and checked against the source page in code. |
| Conditions | Nemotron adapter (checkpoint 400) and its untouched base, each given the same source passage and question; greedy decoding, 512 new tokens, seed 20261006. One run; failures stay in the denominator; no retries or tuning afterward. |
| Scoring | GPT-6 Luna and Claude Sonnet 5.5 score every output blind; I review the main run in full and the cut run where the judges disagree. Pass = correct, supported, no unsupported claim. Citation identifiers are checked in code; on unanswerable questions a correct answer is a decline. |
| Robustness | Each passage cut to its first 60% of characters; questions whose evidence survives the cut and questions whose evidence is removed are scored separately (the second group should be declined). |
| Subgroups | Document family, question type, and answerable versus unanswerable questions. |
| Thresholds | From the PM Plan: +10 points over base on answerable accuracy; at least 95% correct citations; under 2% unsupported claims; at least 90% correct abstention. Safety routing and general-task regression were not tested. |

## 3. Iteration Narrative

The run log (16 training runs, four development experiments and the final test) gives this arc. Failed attempts stay in the record because they ruled things out.

| When | What I tried | Expected | What happened | What I concluded |
|---|---|---|---|---|
| Sep 16–18 | First Muse fine-tunes | Adapter beats base | Used earlier data-generation and training methods that I later amended | Excluded the runs; all later runs use the amended data, split and procedure. |
| Sep 29–Oct 1 | 16 runs: 4 families × 200 / 509 / 750 / 1,600 records, same validation set | More data lowers validation loss | True for three families; Muse flat after 509. From 509 records up, every family overfit after 1–2 epochs. Early Nemotron and Ornith attempts failed (one on an unsupported warm-up setting) | Use 1,600 records and the early checkpoint. Fixed the training runner and reran all eight runs from scratch. |
| Oct 6–7 | Week 8 error review: 15 questions × base and adapter (Nemotron) | Adapter more accurate | Strict accuracy 2/15 → 4/15; 0 of 30 citations correct; invented rules most common | Citations are format only; question-only input invites recall errors. |
| Oct 7 | Refinement: added an instruction to use only verifiable source facts (8 questions, 2 documents) | Fewer unsupported claims | Base swapped answers for “cannot verify” (correct 3/8 → 0/8); adapter unchanged at 4/8 and still cited wrong sources on 8/8 | Prompt wording alone does not fix grounding. Not promoted. |
| Oct 8 | Cross-family probe: 192 outputs scored by a local AI judge | A fair comparison | The first judge counted refusals as supported answers, so I withdrew it; the corrected judge requires a real answer | Three adapters tie at 4/24; selected Nemotron (Milestone 1). |
| Oct 14 | Changed the final-test input to supply the source passage | Isolates what fine-tuning adds | Evaluation harness built; passed separate specification and code-quality reviews | Ready once the fresh benchmark is locked. |
| Oct 19 | One-time final test: 150 fresh questions, base vs. adapter, passage supplied | Adapter at least 10 points better | Adapter -2.5 points vs. base on answerable questions; declined 2 of 30 unanswerable questions (base 24) | Not promoted. Training only on answerable questions taught the model never to decline. |

What the failures ruled out: the early data methods, longer training (more epochs overfit), prompt-only fixes, refusal-friendly scoring and, now, the idea that this adapter adds value once the source is supplied. The one thing I would change is the training data: it contained no unanswerable examples, and the test shows the consequence. A retrained adapter would need its own fresh test; this one cannot be reused.

## 4. Test Results (Final Test, October 19)

Table 1 reports the single final test: the same 150 questions, settings and metrics for the untouched base and the adapter, against the thresholds in my PM Plan. Answerable and unanswerable questions are reported separately so a pooled score cannot hide an abstention failure. Intervals are 95% Wilson intervals.

**Table 1. Baseline, model and threshold on the final test**

| Measure | Untouched base | Adapter (model) | Threshold | Final-test verdict |
|---|---|---|---|---|
| Strict accuracy, 120 answerable questions | 114/120 (95.0%); CI 90–98% | 111/120 (92.5%); CI 86–96% | +10 points over base | Not met (-2.5 points; p = 0.508) |
| Answers with an unsupported claim (lower is better), 150 questions | 6/150 (4.0%) | 25/150 (16.7%) | Under 2% | Not met (both) |
| Answer names the correct document and page, 120 answerable questions | 2/120 (1.7%) | 11/120 (9.2%) | At least 95% | Not met (both) |
| Correct abstention, 30 unanswerable questions | 24/30 (80.0%); CI 63–90% | 2/30 (6.7%); CI 2–21% | At least 90% | Not met (adapter far below; base 80%) |
| Safety routing; general-task regression | Not tested | Not tested | 95%; within 3 points | Not tested: no safety or general-task items |

**Table 2. Confusion-matrix equivalent: outcome of every question (number of questions)**

| Outcome | Untouched base | Adapter |
|---|---|---|
| Answerable (120): answered, correct and fully supported | 114 | 111 |
| Answerable: answered, but wrong, incomplete or with an unsupported claim | 6 | 9 |
| Answerable: declined although the passage had the answer | 0 | 0 |
| Unanswerable (30): correctly declined without inventing a fact | 24 | 2 |
| Unanswerable: answered anyway, or declined while inventing a fact | 6 | 28 |
| Generation failed or cut off at the 512-token limit (main run) | 0 | 0 |

**Business units.** Per 100 questions whose answer is in the source, the base gives about 95 answers an analyst could use as drafted and the adapter about 93; neither ever declined an answerable question. Per 100 questions whose answer is not in the source, the base would give a confident answer about 20 times and the adapter about 93 times, and each of those is a policy statement the source does not support. Median answer time was 1.1 seconds for the adapter and 0.8 for the base. Manual lookup takes about 84 seconds per simple question, but an analyst must still check every draft, and the adapter adds no accuracy over the base to pay for that effort, so the result supports no time-saving claim for the adapter.

## 5. Error and Robustness Analysis

Table 3 gives each of the 150 main-run answers one label for its main problem, so every column adds up to 150. The development columns (my Week 8 review and the AI judge's probe labels) are in Appendix A.

**Table 3. Main problem in each answer, final test (number of answers)**

| Main problem in the answer | Untouched base | Adapter | What would fix it |
|---|---|---|---|
| No error | 138 | 113 | — |
| Wrong or incomplete answer | 6 | 8 | Show the quoted source value beside the answer; check effective dates and editions. |
| Right answer plus an unsupported extra claim | 0 | 1 | Insert the citation and quote from the passage instead of letting the model add detail. |
| Declined although the source had the answer | 0 | 0 | Teach when to decline, using unanswerable training examples. |
| Answered or invented when the source had no answer | 6 | 28 | Add unanswerable and "not in this page" examples to training; check that a quote supports the answer. |
| Total answers checked | 150 | 150 | |

Fewer is better in every row except "No error". Three patterns stand out. First, the adapter's dominant failure is answering when it should decline: it answered 28 of 30 questions whose answer was missing, as I predicted from Milestone 1: every training question had an answer. On answerable questions the two models are close (the base missed 6 and the adapter 9 of 120). Second, citations are

almost absent: only 4 base and 13 adapter answers named a document and page the code could check, and of those the adapter was right on 11. The 95% threshold was out of reach because the models rarely cite at all. Third, declining was never a problem on answerable questions: neither model declined one.

**Check on the measurement itself.** The two judges rated all 300 main-run outputs. They gave the same pass or fail on 292 of 300 and differed on at least one rating for 9; I decided those, siding with Luna on 5 and Sonnet on 4, and I changed no agreed rating. On the robustness run they differed on 17 of 300 outputs and I decided those too (12 Luna, 5 Sonnet). The conclusion does not depend on my calls: scored by Luna alone the answerable pass counts are 114 (base) and 112 (adapter) of 120, and by Sonnet alone 116 and 113. Because I recorded approval once and flagged nothing, my review added no corrections. It confirms that no rating I read looked wrong; it is not evidence that the judges are error-free. In development, the AI judge and I agreed on 12 of 12 for correctness, support and the overall pass, but on citation validity in only 3 of 12, which is why code checks citations.

**Table 4. Robustness: passage cut to its first 60% (number of questions)**

| Measure | Untouched base | Adapter | How to read it |
|---|---|---|---|
| Evidence survives the cut (66 questions): strict pass, full passage → cut passage | 62 → 64 (+3.1 points) | 59 → 62 (+4.5 points) | Small changes of 2–3 questions; within chance, not an improvement. |
| Evidence removed by the cut (54 questions): correctly declined | 23/54 (42.6%) | 0/54 (0.0%) | Expected behavior is to decline. |
| Unanswerable questions, cut run (30): correctly declined | 25/30 | 2/30 | Same pattern as the main run. |

The cut-passage run tests a realistic loss of context. When the evidence survives, both models keep their accuracy. When the evidence is removed, neither behaves safely: the base declined 43% of the time and the adapter never (0 of 54). Cut-run answers were longer; 6 adapter answers hit the 512-token limit and were scored as returned (none for the base). This evaluates the frozen model and does not reopen tuning. Earlier, on October 7, I reran 8 development questions with a reworded instruction: the adapter's correct answers stayed at 4/8 and the base fell from 3/8 to 0/8 as it switched to refusing. That is weak evidence, given the small sample.

## 6. Subgroup Performance

The segments my decision affects are the document types an analyst consults, the kind of question asked, and answerable versus unanswerable questions. Table 5 gives main-run counts. Cells are small, so differences of one or two questions are within chance.

**Table 5. Strict-pass counts by group (passes out of questions in the group)**

| Group | Answerable: base | Answerable: adapter | Unanswerable: base | Unanswerable: adapter |
|---|---|---|---|---|
| Administrative Code chapters | 38/40 | 38/40 | 5/9 | 2/9 |
| DME prior-authorization checklist | 8/8 | 8/8 | 2/2 | 0/2 |
| Waiver program documents | 38/40 | 37/40 | 8/10 | 0/10 |
| Proposed rule notice | 6/6 | 6/6 | 2/2 | 0/2 |
| Provider alerts | 24/26 | 22/26 | 7/7 | 0/7 |
| Value lookup questions | 37/38 | 36/38 | — | — |
| Date or edition questions | 30/30 | 30/30 | — | — |
| Exception questions | 20/22 | 20/22 | — | — |
| Multiple rules questions | 27/30 | 25/30 | — | — |

**How to read Table 5.** Higher is better in every cell. Unanswerable questions were written per document, so question-type rows show only answerable counts. No group shows the adapter ahead of the base on answerable questions. The largest gaps are Provider alerts (24 vs. 22 of 26) and Multiple-rule questions (27 vs. 25 of 30), two questions each, which is within chance. The one real contrast is between answerable and unanswerable questions: the adapter declined correctly in 2 of 9 Administrative Code cases and 0 of 21 others, while the base declined correctly in 5 of 9 Administrative Code cases and 19 of 21 others. Administrative Code chapters are the hardest for the base to decline, so a reviewer should check those answers first.

**NIST AI RMF, tied to these results.** Measure: a 16.7% unsupported-claim rate and a 6.7% correct-abstention rate fail the framework's "valid and reliable" characteristic, and answers that rarely name a page fail "accountable and transparent", because a reviewer cannot trace the answer to its rule. Manage: these results trigger the deployment block in my plan, so the model stays in testing. Govern: an analyst must approve every answer, and a named owner tracks manual editions. Map: the harm is wrong billing guidance reaching providers, which can delay or deny services to Medicaid members.

**EU AI Act.** The system is used in Alabama, so the Act does not apply directly. If it did, a tool that evaluates people's eligibility for public benefits would be high-risk (Annex III). My scope excludes eligibility decisions and keeps a human reviewer, which should keep it outside that category, though transparency duties for AI-generated text could still apply.

## 7. Limitations and Deployment Readiness

- **Easy answerable questions.** With the passage supplied, the base scored 95%, so a gain of 10 points was arithmetically impossible (the maximum was 5). The test cannot show a smaller real benefit, and it measures reading a supplied page, not finding the right page. The unanswerable questions are where the models differed.
- **Benchmark origin.** The questions were written by an AI model (GPT-6.1 Sol), not by me, and checked against the source in code. They come from ten documents, so results may not carry to other manuals.
- **Judging.** Two LLM judges scored the outputs and I reviewed them, but I corrected nothing, so shared judge blind spots could remain. Each judge alone gives the same conclusion. Temperature cannot be set, so repeating the scoring could change a few ratings. With 30 unanswerable questions the abstention intervals are wide (adapter 2–21%, base 63–90%), though they do not overlap.
- **Population and time validity.** Results cover Alabama Medicaid's public documents as downloaded in September and October 2026. The manuals are reissued in dated editions, so an adapter trained on one edition goes stale when the next appears. This is a single run with one seed.
- **Remaining failure modes.** Answering when the source has no answer, very few citations, and (for the adapter) longer, sometimes truncated answers when context is cut (Section 5).
- **Operational requirements.** One GPU with about 64 GB available for the model (63.2 GB allocated after loading on a 96 GB RTX PRO 6000 Blackwell); median answer time 1.1 seconds, 90th percentile 3.2 seconds; the source passage passed in with each question; analyst approval before any answer is used; and an audit log of question, passage, answer and reviewer decision.

**Table 6. Proposed monitoring plan for any pilot**

| What I would monitor | How | Trigger and response |
|---|---|---|
| Unsupported-claim rate | An analyst checks a random 25 answers each week | Above 2% two weeks running: stop drafting and re-evaluate |
| Citation correctness | Same weekly sample, checked against the cited page | Below 95%: switch to inserted citations and re-evaluate |
| Abstention | Five seeded unanswerable questions each week | Fewer than 90% declined: block use |
| Source editions | Compare manual edition dates with the training snapshot monthly | New edition of a covered chapter: re-run the evaluation; retrain only if accuracy drops |
| Latency | Log the answer time of every request | 90th percentile above 10 seconds: capacity review |

**Readiness: not ready.** A pilot requires passing every threshold on a final test. The adapter fails four of four measured thresholds. The untouched base with the passage supplied performs better on abstention and unsupported claims, yet still misses the abstention and citation thresholds, so it is not ready either. Autonomous policy decisions stay out of scope even then.

## 8. Repository

Course repository (public): github.com/porespellar/INAI6986_project_deliverables

Final-test notebook: notebook/final_test_replay_20261019.ipynb

The public notebook recomputes the reported metrics from 600 de-identified final-rating records and a scrubbed result summary. It excludes benchmark questions and keys, source passages and page locators, model answer text, raw judge requests and responses, and the signed review spreadsheet. It does not rerun the model or judges and cannot independently recheck answer correctness or source support. The private repository retains the complete audit record. The development notebook (current_evidence_replay_20261018.ipynb) reproduces the Appendix A figures.

## AI Use Disclosure

I used Hermes Agent with GPT-6 Luna and GPT-6.1 Sol (October 1–16) and Claude Opus 5.5, Claude Sonnet 5.5 and GPT-6.1 Sol (October 18–19) to draft and revise this report and write its evaluation, analysis and build code. GPT-6.1 Sol wrote the final-test questions; GPT-6 Luna and Claude Sonnet 5.5 scored the outputs and neither wrote questions. Local Qwen models generated training records and historical advisory scores. I designed the evaluation, chose the candidate, thresholds and judging plan, made the Week 8 ratings, reviewed the judges' final-test ratings, checked the reported figures against the notebook output and remain responsible for the content.

# Appendix A — Development Results (Question-Only Input)

These results precede the final test. Same 24 validation questions, question-only input, greedy decoding, AI-judged. Failed generations count as not correct. Paired exact McNemar tests compare each adapter with its own base.

| Model | Condition | Fully supported (95% CI) | Unsupported claims | Abstained | Failed | Paired p |
|---|---|---|---|---|---|---|
| Nemotron | Untouched base | 4/24 (7%–36%) | 17/24 | 3 | 0 | |
| Nemotron | Adapter | 4/24 (7%–36%) | 19/24 | 0 | 0 | 1.0 |
| Gemma | Untouched base | 0/24 (0%–14%) | 10/24 | 14 | 0 | |
| Gemma | Adapter | 4/24 (7%–36%) | 18/23 | 0 | 1 | 0.125 |
| Ornith | Untouched base | 1/24 (1%–20%) | 0/24 | 23 | 0 | |
| Ornith | Adapter | 4/24 (7%–36%) | 18/23 | 0 | 1 | 0.375 |
| Muse | Untouched base | 0/24 (0%–14%) | 0/8 | 8 | 16 | |
| Muse | Adapter | 0/24 (0%–14%) | — | — | 24 | 1.0 |

**Table A2. Development error labels (number of answers): my Week 8 review (15 questions) and the AI judge (24)**

| Main problem | My review: base | My review: adapter | AI judge: base | AI judge: adapter |
|---|---|---|---|---|
| Invented rule | 9 | 10 | 11 | 4 |
| Wrong value | 3 | 0 | 6 | 15 |
| Wrong citation | 3 | 5 | 0 | 2 |
| Declined though the source had the answer | 0 | 0 | 3 | 0 |
| Answered without the needed evidence | 0 | 0 | 0 | 1 |
| No error | 0 | 0 | 4 | 2 |

# Appendix B — Evidence Files

| Claim | File in the source-of-record repository |
|---|---|
| Final-test protocol, sealed before the first output | artifacts/milestone2_completion/final_test_20261019/PROTOCOL.md; SEAL_SHA256SUMS.txt |
| Decisions recorded before authoring (benchmark author, judges, review scope) | artifacts/milestone2_completion/final_test_20261019/DECISIONS_20261019.md |
| Benchmark (key, model inputs, verification) | artifacts/milestone2_completion/final_test_20261019/benchmark/ |
| 600 raw outputs | artifacts/milestone2_completion/final_test_20261019/outputs/outputs.jsonl |
| Both judges' raw requests, responses and scores | artifacts/milestone2_completion/final_test_20261019/judging/ |
| My review decisions (26 calls, 0 flags) and sign-off | artifacts/milestone2_completion/final_test_20261019/review/ (user review decisions and sign-off); AMENDMENT_1_review_sheet.md |
| Final ratings and every final-test number | artifacts/milestone2_completion/final_test_20261019/results/final_ratings.jsonl; final_results.json |
| Development: probe, 192 outcomes, Week 8 review, refinement | artifacts/milestone2_completion/development/ and review/batch_b/ |
| All derived development numbers | artifacts/milestone2_completion/class_package_20261018/ development_analysis_20261018.json |